INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

GRC205 Incident Management and Business Continuity, Week 21

Business Continuity and Disaster Recovery Assignment

Northbridge Skills Institute: Main Site Flood and Power Failure

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All organisations, people, roles, contact details and figures in this report are fictional exercise data. Results from the drill timings are modelled baseline results, not a live test.

Introduction

This report sets out the responsibilities of the Business Continuity and Disaster Recovery (BC/DR) Coordinator for Northbridge Skills Institute following the flooding and power failure that closed the main site at 08:00 WAT on Day 0, a Monday. All times are West Africa Time. The case uses fictional organisational details. Assumptions are labelled A1 onwards and proposed controls are labelled P1 onwards, with registers in Appendices A and B. The workings underlying the impact estimates, objectives and recovery schedule appear in the main tables and Appendix C. Drill outcomes are modelled baseline results, not observations from a live recovery test. [1]

Task 1 Business Impact Analysis

1.1 Five-function BIA

The BIA follows six phases: identify critical functions, determine impacts, establish RTO and RPO, identify resources and prioritise recovery. Table 1 records each function. All five functions share the same people (90 staff and two technicians), premises (the closed main site and an alternate facility with 25 workstations from 10:00), suppliers (Microsoft 365, the payment provider, the bank, the offsite backup provider and the alternate facility provider) and information (DB01 records, AUTH01 accounts and FS01 files). Owner roles are fictional (A2). [2, slide 5]; [3, Sec. 3.2].

These shared needs drive the minimum services. People are the binding constraint, because the same two technicians restore every system while business coordinators run the manual processes. Premises matter mainly for F1, since only the examination needs seats in a controlled room. Suppliers become critical during the outage: the payment provider and the bank hold the independent records that make later reconciliation possible. Information needs differ by function. F2 and F4 need complete and accurate records more than speed, whereas F1 and F3 need timely access before their deadlines.

Table 1. Business Impact Analysis for F1 to F5 (owner roles fictional, A2)

Function / priority	Business profile and dependencies	Disruption impact and continuity service
F1 Teaching and assessment (LMS01, DB01) Priority 1	**Owner:** Head of Assessment **Users:** 1,800 learners; 120 exam candidates (40 need a centre workstation, 80 can use own devices); tutors **Resources:** Tutors, invigilators, learner support; alternate facility workstations; Microsoft 365; LMS01 data, question bank, adjustments list **Dependencies:** Recovery network, AUTH01, DB01, LMS01; staff; secure access; communication	**Impact over time:** 0-3 h: exam preparation disrupted. 11:00: exam cannot start, NGN 120,000 rescheduling and learner distress. Beyond 24 h: teaching and coursework deadlines slip. **Minimum acceptable service:** Exam postponed by a recorded decision and re-run on validated LMS01 with an approved paper variant, invigilators and signed attendance registers; teaching notices through Microsoft 365.
F2 Fees and finance (FIN01, DB01) Priority 2	**Owner:** Bursar **Users:** Fee-paying learners; finance staff **Resources:** Two finance officers; payment provider portal; bank; FIN01 receipts and wallet records **Dependencies:** Recovery network, AUTH01, DB01, FIN01; payment provider	**Impact over time:** Fee confirmation due 16:00 Day 0. Unconfirmed learners and disputes build each hour; the reconciliation backlog grows. **Minimum acceptable service:** Payments confirmed from the provider portal; provisional confirmations from the finance mailbox; numbered manual register with two-person approval; no refunds or wallet changes until FIN01 is reconciled.
F3 Admissions and registration (REG01, DB01) Priority 3	**Owner:** Registrar **Users:** Applicants; learners changing registration; admissions staff **Resources:** Two admissions officers; Microsoft 365 forms and mailbox; REG01 records **Dependencies:** Recovery network, AUTH01, DB01, REG01	**Impact over time:** Admissions close 18:00 Day 0. Applications cannot be entered; applicants may be lost and reputation harmed. **Minimum acceptable service:** Timestamped online intake with automatic reference numbers; intake log; later entry into REG01 with a second-person check.
F4 Staff payroll (HR01, DB01) Priority 4	**Owner:** HR Manager **Users:** 90 staff **Resources:** Payroll officer; Bursar as approver; bank portal; HR01 records **Dependencies:** Recovery network, AUTH01, DB01, HR01; bank	**Impact over time:** Low on Day 0. Bank deadline 12:00 Day 1. Incorrect or duplicate instructions harm staff and need manual correction. **Minimum acceptable service:** Payroll file built only from accepted HR01 data, compared with the last approved run, approved by two people and submitted once.
F5 Records and certificates (FS01) Priority 5	**Owner:** Records Manager **Users:** Learners, graduates and employers requesting records **Resources:** Records officer; signing authority; FS01 files **Dependencies:** Recovery network, AUTH01, FS01	**Impact over time:** Low on Day 0. The 48-hour commitment is breached after 08:00 Day 2. Integrity or access failures are severe at any time. **Minimum acceptable service:** Requests logged in a restricted mailbox and acknowledged; documents issued only from integrity-checked FS01 files with Registrar approval.

1.2 Worked impact calculations

Each function's cost is its hourly rate multiplied by the hours of disruption, applied for the first 24 hours only, with no recovery or continuity measures. The examination trigger applies in both windows because 11:00 is three hours after the disruption, so the NGN 120,000 is added once to F1. Table 2 shows the working. The totals are NGN 1,200,000 for 8 hours and NGN 3,360,000 for 24 hours. These are exercise estimates, not observed losses. [1, p. 2].

Table 2. Disruption cost without recovery (rate x hours + one-off trigger)

Field	Information
F1	
Rate NGN/h	60,000
8-hour working	60,000 x 8 = 480,000; + 120,000 trigger
8-hour cost NGN	600,000
24-hour working	60,000 x 24 = 1,440,000; + 120,000 trigger
24-hour cost NGN	1,560,000
F2	
Rate NGN/h	40,000
8-hour working	40,000 x 8
8-hour cost NGN	320,000
24-hour working	40,000 x 24
24-hour cost NGN	960,000
F3	
Rate NGN/h	20,000
8-hour working	20,000 x 8
8-hour cost NGN	160,000
24-hour working	20,000 x 24
24-hour cost NGN	480,000
F4	
Rate NGN/h	10,000
8-hour working	10,000 x 8
8-hour cost NGN	80,000
24-hour working	10,000 x 24
24-hour cost NGN	240,000
F5	
Rate NGN/h	5,000
8-hour working	5,000 x 8
8-hour cost NGN	40,000
24-hour working	5,000 x 24
24-hour cost NGN	120,000
Combined	
Rate NGN/h	135,000
8-hour working	135,000 x 8 = 1,080,000; + 120,000
8-hour cost NGN	1,200,000
24-hour working	135,000 x 24 = 3,240,000; + 120,000

Field	Information
24-hour cost NGN	3,360,000

The estimates have two main limitations. First, a flat hourly rate ignores timing. The real F2 cost is concentrated around the 16:00 deadline and the F4 cost around 12:00 on Day 1, not spread evenly. Second, the rates are treated as separate and stop at 24 hours, although the outage lasts 72 hours, and they leave out the cost of reconciliation after recovery and the shared effect of losing DB01.

The naira figures measure financial impact only. Learner welfare (examination anxiety, travel and accessibility), reputation (lost applicants and trust) and data integrity (wrong payroll, missing receipts) are separate impacts. They are judged qualitatively in Table 1 and can outweigh the cost figures.

1.3 Priorities and dependency risks

The business priority follows the deadlines and rates: F1 (11:00 exam, highest rate), F2 (16:00), F3 (18:00), F4 (12:00 Day 1) and F5 (48 hours). The technical restoration order is different, because F1 to F4 all need the recovery network, AUTH01, the security checks and DB01 first. Task 5 shows that the 225-minute chain to LMS01 cannot finish before 11:45. No restoration can therefore save the 11:00 exam, so F1's top priority leads to an early continuity decision rather than a faster restore. FS01 is restored early because the second technician is otherwise idle, not because F5 is urgent. Table 3 lists the dependency risks. [1, pp. 2, 4].

Table 3. Important dependency risks

Field	Information
D1	
Dependency risk	DB01 is the single data base for F1 to F4
Effect	One corrupt or failed restore stops four functions
Treatment	Verified restore chain, 30-minute incrementals and monthly restore tests (Table 6)
D2	
Dependency risk	AUTH01 snapshot is 8 hours old
Effect	Accounts, passwords and permission changes made after 00:00 are lost, which affects access to every function
Treatment	Two-hourly snapshots; re-apply logged changes after restoration
D3	
Dependency risk	Production and backup administration share one identity
Effect	One compromised account can damage both production and backups
Treatment	Separate identities with MFA; break-glass account (CA1)
D4	
Dependency risk	The offsite provider's access-independence controls are undocumented
Effect	Backups may be reachable by the same attacker or unavailable when needed
Treatment	Written attestation and a witnessed test (CA4)
D5	
Dependency risk	Only two technicians
Effect	Sickness or fatigue delays every restore
Treatment	Cross-train a third person (CA8)
D6	
Dependency risk	25 workstations for 40 centre learners
Effect	The exam cannot run in one sitting
Treatment	Split sittings (Table 14); surge contract (CA5)

Task 2 Recovery objectives and backup design

2.1 Definitions and targets

The Recovery Time Objective (RTO) is the maximum time a function or system can remain unavailable before the impact becomes unacceptable. The Recovery Point Objective (RPO) is the maximum acceptable data loss, measured in time before the disruption. The Maximum Tolerable Downtime (MTD) is the total outage the owner is willing to accept, taking all impacts into account. NIST states that the RTO must normally be shorter than the MTD, because reprocessing time is added after recovery. The usual definition measures RTO up to full restoration. In this report, recovery is declared when the minimum acceptable service in Table 1 is accepted by the function owner, (P1). Full restoration then follows through the controlled return described in Section 4.6. [2, slide 6]; [3, Sec. 3.2].

Table 4 sets the targets (P2). Each MTD comes from a stated deadline. Each RTO leaves a buffer for business acceptance, reconciliation and delay. The F2 buffer is the smallest, at three hours.

These objectives are management proposals, not verified technical service-level agreements. The distinction matters because the exercise gives only an estimated time for the first critical service under each package, not an end-to-end recovery guarantee for every function. The MTD is the point at which wider business harm becomes intolerable; it must be longer than the operational RTO to allow time for business verification, reprocessing and the resolution of unexpected errors. The MTD for F2 is tied to the 16:00 payment-confirmation deadline, while payroll has more flexibility because the bank submission is not due until the following day.

Table 4. Recovery objective matrix (targets are proposals, P2)

Field	Information
F1	
RTO	5 h
RPO	1 h
MTD	24 h
Buffer (MTD - RTO)	19 h
Justification	The 11:00 exam (3 h) is handled by rescheduling, which no RTO can meet. LMS01 is needed for teaching and for the Day 1 sitting. The 5 h RTO matches the upper warm estimate.
F2	
RTO	5 h
RPO	1 h
MTD	8 h
Buffer (MTD - RTO)	3 h
Justification	The 16:00 confirmation deadline is 8 h after the disruption. The buffer covers acceptance and reconciliation before 16:00.
F3	
RTO	6 h
RPO	1 h
MTD	10 h
Buffer (MTD - RTO)	4 h
Justification	Admissions close at 18:00, 10 h after the disruption. The buffer allows online-intake records to be keyed in before closing.

Field	Information
F4	
RTO	12 h
RPO	4 h
MTD	24 h
Buffer (MTD - RTO)	12 h
Justification	The bank deadline is 28 h away; 4 h is reserved for checks and two-person approval, giving an MTD of 24 h. Payroll data changes slowly.
F5	
RTO	24 h
RPO	4 h
MTD	48 h
Buffer (MTD - RTO)	24 h
Justification	The 48-hour service commitment sets the MTD; file integrity matters more than speed.
AUTH01 (support)	
RTO	Before DB01
RPO	4 h
MTD	As F1-F3
Buffer (MTD - RTO)	-
Justification	A prerequisite for every function, so it is restored first.

2.2 Checkpoint analysis

Table 5 measures each checkpoint's age from the 08:00 disruption. DB01 meets the 1-hour RPO for F1 to F3, but with no margin. AUTH01 exceeds its 4-hour RPO by 4 hours, and FS01 exceeds its 4-hour RPO by 5 hours. [1, p. 3].

Table 5. Age of the latest recoverable checkpoints at 08:00 Day 0

Field	Information
B01 + B02 (DB01)	
Checkpoint	07:00 Day 0
Working	08:00 - 07:00
Age	1 h
Functions	F1, F2, F3, F4
RPO	1 h (F1-F3); 4 h (F4)
Gap	Meets, zero margin for F1-F3
B03 AUTH01	
Checkpoint	00:00 Day 0
Working	08:00 - 00:00
Age	8 h
Functions	All (access)
RPO	4 h
Gap	4 h over
B04 FS01	
Checkpoint	23:00 Day -1

Field	Information
Working	24:00 - 23:00 + 8 h
Age	9 h
Functions	F5
RPO	4 h
Gap	5 h over
Disconnected full	
Checkpoint	17:00 Day -3
Working	7 h + 48 h + 8 h
Age	63 h
Functions	Fallback (A10)
RPO	-
Gap	Last resort only

Some data can be rebuilt from independent records. Payments can be reconciled from the payment provider's export. Applications and requests made after 08:00 are held in the temporary intake logs, and payroll can be checked against the last approved run and the approval emails. Other data cannot be assumed recoverable: DB01 transactions entered between 07:00 and 08:00 (receipts, wallet changes, registration changes and LMS submissions), AUTH01 changes made after 00:00, and FS01 files created after 23:00 on Day -1. Owners must ask users to re-submit these items and record each exception.

Reconciliation must distinguish events completed before the last checkpoint from actions performed during the outage. Each temporary entry should carry a unique reference, timestamp, creator and approver, plus a reference to any source evidence such as a payment-provider transaction identifier. After restoration, owners compare these records against the recovered databases before accepting changes. A missing record is not automatically a failed backup: it may have been created after 07:00, when the DB01 checkpoint was taken. Conversely, a duplicate or unexplained alteration is not dismissed as ordinary data loss. It is escalated for further investigation, with supporting logs preserved.

2.3 Backup and restoration design

Table 6 addresses the RPO gaps using the 3-2-1 principle: three copies, two storage media and at least one offsite copy, with an additional independently protected offline or immutable copy (P3). To restore DB01, select the latest verified weekly full, apply the most recent valid differential made after that full when applicable, then apply every subsequent incremental in sequence until the chosen checkpoint. For the specific Monday 08:00 disruption, the proposed chain is a Sunday 23:00 full followed by 17 half-hour incrementals from 23:30 to 07:30. The 07:30 recovery point is 30 minutes old and therefore inside the one-hour proposed RPO. Appendix C records the arithmetic. [2, slide 8]; [3, Sec. 3.4]

Table 6. Backup schedule and protection design (proposals, P3)

Field	Information
DB01	
Full	Weekly, Sunday 23:00
Incremental or differential	Differential daily at 23:00 (Mon-Sat); incremental every 30 min
Retention	Incrementals 14 days; differentials 35 days; weekly fulls 13 weeks; month-end fulls 12 months
Placement (3-2-1)	Copy 1 production; copy 2 encrypted disk at the offsite provider; copy 3 immutable (35-day lock) or offline copy at a second, independent location

Field	Information
Protection	Encryption in transit and at rest; keys held apart from the provider; separate backup identities with MFA; no shared production identity
Restore test	Monthly restore into an isolated environment
AUTH01	
Full	Weekly system-state backup
Incremental or differential	Snapshot every 2 h and before any administrative change
Retention	30 days; month-end 12 months
Placement (3-2-1)	As DB01
Protection	As DB01; break-glass account sealed
Restore test	Quarterly
FS01	
Full	Weekly, Sunday 23:00
Incremental or differential	Incremental every 2 h
Retention	Incrementals 30 days; fulls 13 weeks; month-end 12 months
Placement (3-2-1)	As DB01
Protection	As DB01; access lists preserved
Restore test	Quarterly, plus file hash sampling

Neither offsite storage nor a live replica proves recoverability on its own. An offsite copy can be incomplete, corrupt, encrypted with unavailable keys, or reachable through the same compromised identity. A replica copies corruption and malicious deletions almost immediately. NIST advises testing backups regularly so that files can be retrieved without errors or lost data. Only a successful restore followed by application and data validation proves that a backup can be recovered. Table 7 defines the checks. For the baseline recovery drill, B01 to B04 are assumed to pass their integrity checks. The provider's access-independence remains unverified (A6, CA4). [3, Secs. 3.4, 4].

Table 7. Verification of backup integrity, access independence and usability

Field	Information
Integrity	
Method	Checksum each backup job; confirm the catalogue shows an unbroken chain to the target point
Evidence	Job report with verified hashes; signed chain list
Frequency	Every job; before any restore
Access independence	
Method	Separate backup identities with MFA; provider attestation; review of backup-console logons
Evidence	Access review; signed attestation; logon report
Frequency	Monthly; quarterly attestation
Usability	
Method	Restore to an isolated environment; database consistency check; test login; sample records compared with source reports; last transaction time
Evidence	Signed restore test record
Frequency	Monthly (DB01); quarterly (AUTH01, FS01)

Task 3 Recovery strategy and budget

3.1 Option comparison

Table 8 compares the packages against the targets in Table 4. The quoted windows are supplier planning estimates, not guarantees. The general site characteristics follow the standard descriptions of cold, warm and hot sites. [1, p. 3]; [2, slide 7].

Geography, staffing and supplier dependency separate the options as much as speed does. None of the packages states where its recovery site is, so all three carry the same unconfirmed risk that the site sits in the same flood area or on the same power supply (A6). Cold would ask two technicians to build and configure systems under time pressure. Hot moves the work to continuous replication, which the provider must run correctly every day. Warm restores from backups that Northbridge controls, so a verified restore point can be chosen before data is loaded. This matters for propagation risk, because replication copies corruption or malicious changes as quickly as it copies good data.

Table 8. Option matrix (supplier figures are estimates)

Field	Information
Estimated first critical service	
Cold	24-48 h
Warm	3-5 h
Hot	About 1 h
Against RTOs (5 h F1-F2, 6 h F3)	
Cold	Fails F1-F3 and the F2 and F3 MTDs
Warm	Meets them at the upper estimate only
Hot	Meets them, and may meet the 11:00 exam if validated
Data-loss window against 1 h RPO	
Cold	Up to 24 h: fails
Warm	1-6 h: fails unless Northbridge's own 30-minute chain is used
Hot	Up to 15 min: meets
Staffing (two technicians)	
Cold	Heavy build effort; slowest
Warm	Feasible: modelled drill completes in 4 h 15 min
Hot	Lower recovery effort, but continuous monitoring is needed
Geography	
Cold	Location unknown (A6)
Warm	Location unknown (A6)
Hot	Location unknown (A6)
Supplier dependency	
Cold	Low during normal operation; high at recovery
Warm	Moderate
Hot	High and continuous
Propagation of corruption or malicious change	
Cold	Low (offline restore)
Warm	Moderate (restore from a verified point)
Hot	High (near-real-time replication copies changes)

3.2 Budget

Table 9. Itemised annual cost of each package (NGN)

Field	Information
Cold	
Base	600,000
Independent protected backups	400,000
Training and tests	200,000
Total	1,200,000
Limit 3,000,000 minus total	1,800,000
Within limit	Yes
Warm	
Base	1,500,000
Independent protected backups	400,000
Training and tests	200,000
Total	2,100,000
Limit 3,000,000 minus total	900,000
Within limit	Yes
Hot	
Base	2,400,000
Independent protected backups	400,000
Training and tests	200,000
Total	3,000,000
Limit 3,000,000 minus total	0
Within limit	Yes, exactly at the limit with no contingency

3.3 Recommendation, residual risk and investment reasoning

I recommend the Warm package with both required additions, at NGN 2,100,000 a year. This leaves NGN 900,000 (30%) for contingency and price changes. Cold cannot meet the F2 and F3 deadlines. Hot sits exactly at the limit, so any cost increase would breach it, and its near-real-time replication would copy the kind of corruption tested in inject 3. Warm cannot meet the 11:00 exam, which falls only three hours after the disruption, so it relies on the rescheduling procedure. Its 1 to 6-hour data-loss estimate also cannot meet the 1-hour RPO. That RPO therefore depends on Northbridge's own 30-minute backup chain, funded by the NGN 400,000 addition. Table 10 records the residual risks. The Principal accepts them on behalf of the Executive Steering Committee. [1, p. 3].

Table 10. Residual risks and acceptance

Residual risk	Treatment	Accepted by
Rescheduling cost (NGN 120,000) whenever an outage falls on an exam morning	Exam contingency procedure (CA6)	Principal, on advice of the Head of Assessment
RPO depends on Northbridge's own backup chain, not the package	30-minute incrementals with integrity checks (CA3)	Principal, on advice of the ICT Manager
Supplier timings are unvalidated	Contract test clause before payment	Principal and Bursar
Two-technician dependency	Cross-training (CA8)	Principal
Workstation shortfall (25 for 40 learners)	Surge contract (CA5)	Principal

Compared with the impact figures, Warm costs less than one 24-hour unrecovered outage: NGN 2,100,000 against NGN 3,360,000, or 62.5%. If hourly costs stop at the modelled declared recovery times (A16), a 24-hour event would cost about NGN 692,500 with Warm, which avoids NGN 2,667,500 (Appendix C). A single-outage estimate still cannot establish an annual return on investment. Annual value depends on how often such events occur, which is unknown. The rates are estimates, some costs occur anyway (the exam trigger), and the most serious harms are not financial. Before approving the expenditure, the Steering Committee needs evidence: Northbridge's flood and outage history, a witnessed restore of DB01 to the warm site within 5 hours, the supplier's site location and its flood and power exposure, a contract with tested service levels and service credits, the provider's access-independence attestation, and competing quotes.

Task 4 Business continuity and disaster recovery plan

4.1 Activation, roles and coordination

Table 11 sets out the core elements of the plan, following the standard plan structure and NIST's activation criteria. The plan works under a unified command led by the Principal. The continuity team (the BC/DR Coordinator and the function owners) runs the temporary procedures and business acceptance. The technical recovery team (the ICT Manager and two technicians) follows the runbook. Facilities manages the alternate facility and site clearance. The incident response team stays on standby. All teams report through one situation report, and an IR request to pause restoration for evidence preservation is decided and recorded by the Principal. The BC/DR Coordinator keeps the decision log and chairs a short coordination call before each situation report, so that technical progress, temporary procedures and communications stay consistent. [2, slide 10]; [3, Ch. 4].

The incident log should show each decision, the reason for it, the approving authority, the time of approval and the action owner. Even with Microsoft 365 available, the fallback SMS and telephone tree must be practical because mobile connectivity may deteriorate during severe flooding. Printed contact cards should be held by the Principal, coordinator, ICT Manager, Facilities Manager and each function owner. Situation reports should distinguish a technical system that is running from a service that learners and staff can safely use. This avoids premature announcements that an assessment, payment or payroll service has resumed before the business owner signs its acceptance checklist.

Table 11. Plan scope, activation and reporting

Element	Content
Scope	F1-F5, their systems, the main site, the alternate facility and the warm recovery site; all-hazards disruption, with this 72-hour outage as the reference case
Activation thresholds (P4)	Main site, DB01 or AUTH01 unavailable, or expected to be unavailable, for more than 2 hours; an exam within 24 hours under threat; any F2-F4 deadline at risk; any compromise indicator (also activates IR)
Decision maker and deputy	Principal; Deputy Principal as deputy (fictional roles)
Escalation	Coordinator to Principal within 15 minutes of a threshold being met; Principal to Steering Committee if an MTD is at risk or unbudgeted spend exceeds NGN 250,000 (P4); ICT Manager to IR Lead on any compromise indicator
Contact roles	Principal, Deputy Principal, BC/DR Coordinator, ICT Manager, Technicians 1 and 2, function owners, Facilities Manager, Communications Lead, IR Lead, Data Protection Officer, supplier account managers; mailbox bcdr@northbridge.example (fictional)
Alternative communication	Primary: Microsoft 365 email and Teams over mobile internet. Fallback: SMS and voice call tree from staff mobile phones (A13); printed contact cards held by role holders
Situation reporting	09:00, 12:00 and 16:00 on Day 0, then 09:00 and 16:00 daily. Each report gives function status against RTO and MTD, decisions, risks, next actions and communications issued

4.2 RACI matrix

Table 12 assigns exactly one accountable role and at least one responsible role to every activity. The roles reflect the governance arrangements proposed for Northbridge and the agreed responsibility structure. [2, slide 12]

Table 12. RACI matrix. PR Principal; BC BC/DR Coordinator; ICT ICT Manager; TEC Technicians; HA Head of Assessment; FO Owners of F2-F5; FM Facilities Manager; CL Communications Lead; IR IR Lead; DPO Data Protection Officer. A/R = accountable and responsible

Field	Information
1 Plan activation	
Accountable (A)	PR
Responsible (R)	BC
Consulted (C)	ICT, HA, FM
Informed (I)	TEC, FO, CL, IR, DPO
2 Temporary assessment delivery	
Accountable (A)	HA
Responsible (R)	HA, FM
Consulted (C)	BC, ICT, CL, DPO
Informed (I)	PR, TEC
3 Backup selection and validation	
Accountable (A)	ICT
Responsible (R)	TEC
Consulted (C)	IR
Informed (I)	PR, BC

Field	Information
4 System restoration	
Accountable (A)	ICT
Responsible (R)	TEC
Consulted (C)	IR
Informed (I)	PR, BC, HA, FO
5 Business acceptance of restored service	
Accountable (A)	BC
Responsible (R)	HA, FO
Consulted (C)	ICT, TEC
Informed (I)	PR
6 Stakeholder communication	
Accountable (A)	PR
Responsible (R)	CL
Consulted (C)	BC, ICT, HA, FO, IR, DPO
Informed (I)	None
7 Compromise assessment and IR escalation	
Accountable (A)	IR
Responsible (R)	ICT, IR
Consulted (C)	TEC, DPO
Informed (I)	PR, BC, CL
8 Return to primary site and final sign-off	
Accountable (A)	PR
Responsible (R)	BC, ICT, FM
Consulted (C)	TEC, HA, FO, IR
Informed (I)	CL

4.3 Temporary procedures for the 72-hour outage

At a 09:00 go/no-go point (P9), the Principal, advised by the Head of Assessment, postpones the 11:00 examination. The earliest modelled LMS01 restoration is 11:45, before business acceptance, and only 25 workstations exist for the 40 learners who need one. Starting a partly validated exam would put integrity and accessibility at risk, so the NGN 120,000 trigger is accepted (A4). The exam is re-run on Day 1 in two sittings with different approved paper variants (Table 14; A3, A5). Learners with adjustments are seated in the first sitting, with assistive software checked on Day 0. Workstation profiles are wiped between sittings, and business staff work on laptops through Microsoft 365 so that the workstations stay free for learners (A14). Table 13 sets out the procedure for each function. [1, pp. 2, 6].

With 25 centre workstations for 40 centre-dependent learners, the proposed two-sitting plan creates capacity without pretending that the first sitting meets the original 11:00 obligation. Each sitting needs a separate approved examination variant, attendance register, invigilation coverage and an agreed policy for late arrivals or connectivity failure. The 80 own-device candidates require a secure route to the restored portal and a support channel. Learners with disability-related adjustments must be considered when seats are allocated rather than handled after general allocation. Until these controls are approved and tested, the rescheduled examination remains a plan rather than a guaranteed service.

Table 13. Temporary procedures F1 to F5 (staffing numbers are proposals, P5; staff availability A11)

Field	Information
F1	
Temporary procedure	Postpone the exam; teach through Microsoft 365 notices; re-run on Day 1 (Table 14)
Approvals and record controls	Principal approves postponement; Head of Assessment and moderator approve variant B; signed attendance and incident registers
Integrity, privacy and accessibility	Separate variants; invigilation; no personal storage; wiped profiles; adjustments list honoured
Reconciliation	Registers matched to LMS01 submissions; incident log reviewed
Minimum staff, equipment, suppliers	Head of Assessment; 4 invigilators or proctors; 2 learner support staff; 25 workstations; LMS01; facility supplier
F2	
Temporary procedure	Confirm payments in the provider portal; send provisional confirmations; numbered manual register in a restricted Microsoft 365 list; no refunds or wallet changes
Approvals and record controls	Finance officer prepares, Bursar approves; entries timestamped
Integrity, privacy and accessibility	Register restricted to finance; payer data minimised; MFA
Reconciliation	Every provider transaction since 07:00 matched to FIN01; missing receipts posted as approved adjustments; exceptions log
Minimum staff, equipment, suppliers	Bursar; 2 finance officers; 2 laptops; provider portal
F3	
Temporary procedure	Online intake form with automatic reference; registration changes through the admissions mailbox; 18:00 closing judged by timestamp
Approvals and record controls	Intake log; Registrar decides late or unclear cases
Integrity, privacy and accessibility	Restricted form; secure document upload; telephone support for accessibility
Reconciliation	Entries keyed into REG01 with a second-person check; log totals equal REG01 totals
Minimum staff, equipment, suppliers	Registrar; 2 admissions officers; Microsoft 365
F4	
Temporary procedure	No payroll action on Day 0. If HR01 is not accepted by 08:00 Day 1, submit the last approved payroll plus approved variations
Approvals and record controls	HR Manager and Bursar approve; fallback also needs Principal approval; one submission reference
Integrity, privacy and accessibility	Payroll data restricted; secure bank portal
Reconciliation	HR01 compared with the last approved run and logged changes; duplicate check before submission
Minimum staff, equipment, suppliers	HR Manager; payroll officer; Bursar; bank portal
F5	
Temporary procedure	Log requests in a restricted mailbox and acknowledge within 4 working hours (P5)
Approvals and record controls	Registrar approves each issue; issue register

Field	Information
Integrity, privacy and accessibility	Requester identity checked; file hashes compared before issue
Reconciliation	Files changed after 23:00 Day -1 re-checked; lost uploads re-requested
Minimum staff, equipment, suppliers	Records Manager; 1 records officer

Table 14. Baseline Day 1 examination seating (proposal P9; A3, A5, A7)

Field	Information
1	
Time (Day 1)	09:00
Centre learners	25
Workstations used	25 of 25
Own-device learners	80 (remote, A7)
Paper variant	A
Invigilation	2 room invigilators + 2 remote proctors
2	
Time (Day 1)	14:00
Centre learners	15
Workstations used	15 of 25 (10 free for staff)
Own-device learners	0
Paper variant	B
Invigilation	2 room invigilators
Total	
Time (Day 1)	-
Centre learners	40
Workstations used	-
Own-device learners	80
Paper variant	-
Invigilation	120 candidates seated

4.4 DR runbook

Steps 1 to 11 in Table 15 restore technology. Steps 12 and 13 resume an acceptable business service. A restored LMS01 is not an assessment service until the Head of Assessment accepts it.

Table 15. DR runbook

Field	Information
1	
Action	Confirm activation, assign technicians, open recovery log
Prerequisite	Activation decision
Owner	ICT Manager
Completion check	Log open; roles confirmed
Contingency if the step fails	Escalate to BC/DR Coordinator

Field	Information
2	
Action	Select backups: B01 + B02 chain to 07:00, B03, B04; record hashes
Prerequisite	1
Owner	ICT Manager
Completion check	Catalogue shows an unbroken chain; selection signed
Contingency if the step fails	Use the latest verified point; if tampering is suspected, call IR and consider the Day -3 disconnected full (A10)
3	
Action	Recovery network (30 min)
Prerequisite	None
Owner	Technician 1
Completion check	Links to backup source and alternate facility work
Contingency if the step fails	Supplier escalation; alternate link
4	
Action	AUTH01 restoration (30 min)
Prerequisite	3
Owner	Technician 1
Completion check	Directory responds; shared admin identity disabled; new admin credentials issued
Contingency if the step fails	Rebuild essential accounts from documented configuration; IR if anomalies appear
5	
Action	Security and backup checks (30 min)
Prerequisite	4
Owner	Technician 1
Completion check	Restored AUTH01 scanned clean; backup-console logons reviewed; B02 checksums verified
Contingency if the step fails	Halt the affected restore; call IR; choose an earlier verified point
6	
Action	DB01 restoration: B01, then B02 in order to 07:00 (90 min)
Prerequisite	5
Owner	Technician 1
Completion check	Consistency check passes; last transaction about 07:00; record counts match the last reports
Contingency if the step fails	Restore to the last verified incremental; roll back to the pre-apply snapshot
7	
Action	FS01 restoration (60 min)
Prerequisite	4 and 5
Owner	Technician 2
Completion check	Hash sample matches manifest; access lists unchanged
Contingency if the step fails	F5 stays manual; try an alternative copy

Field	Information
8	
Action	LMS01 restoration (45 min)
Prerequisite	6
Owner	Technician 1
Completion check	Login, course page and test submission on a test account
Contingency if the step fails	Roll back the application snapshot; Day 1 exam plan unchanged
9	
Action	FIN01 restoration (30 min)
Prerequisite	6
Owner	Technician 2
Completion check	Test receipt on a test ledger
Contingency if the step fails	Continue the manual register
10	
Action	HR01 restoration (45 min)
Prerequisite	6
Owner	Technician 2
Completion check	Staff count equals the last payroll; no pending batch
Contingency if the step fails	F4 fallback
11	
Action	REG01 restoration (30 min)
Prerequisite	6
Owner	Technician 1
Completion check	Test application record created and removed
Contingency if the step fails	Continue online intake
12	
Action	Business acceptance of each function
Prerequisite	7-11
Owner	Function owner
Completion check	Signed checklist: sample records, last transaction time, access rights; minimum service declared
Contingency if the step fails	Reject; roll back; remain on the manual procedure
13	
Action	Reconcile the gap and temporary records; full backup of the restored state
Prerequisite	12
Owner	Owners; technicians
Completion check	Exceptions closed or escalated; backup verified
Contingency if the step fails	IR if differences are unexplained

4.5 Communication

Table 16 sets out the communication matrix. The initial learner notice follows it.

Table 16. Stakeholder communication matrix

Field	Information
120 exam candidates	
Information need	Postponement, new arrangements, adjustments
Channel (primary / alternative)	Email / SMS (A13)
Owner	Head of Assessment via Comms Lead
Timing	By 09:30 Day 0; arrangements by 18:00
All learners	
Information need	Service status; how to pay, apply and request records
Channel (primary / alternative)	Email / SMS
Owner	Comms Lead
Timing	10:00 Day 0, then daily
Staff	
Information need	Roles, where to report, remote working
Channel (primary / alternative)	Teams / call tree
Owner	BC/DR Coordinator
Timing	08:30 Day 0 and at each situation report
Applicants	
Information need	How to apply before 18:00
Channel (primary / alternative)	Email / SMS
Owner	Registrar
Timing	10:00 Day 0
Payment provider	
Information need	Transaction export; reconciliation support
Channel (primary / alternative)	Email / telephone
Owner	Bursar
Timing	10:00 Day 0 and after FIN01 acceptance
Bank	
Information need	Payroll timing or any delay
Channel (primary / alternative)	Relationship manager
Owner	HR Manager
Timing	By 16:00 Day 0 if at risk
Alternate facility and recovery providers	
Information need	Activation, capacity, extension
Channel (primary / alternative)	Telephone / email

Field	Information
Owner	Facilities Manager; ICT Manager
Timing	08:15 Day 0
Executive Steering Committee	
Information need	Situation reports, decisions, risk acceptance
Channel (primary / alternative)	Teams / email
Owner	Principal
Timing	Each situation report
Data protection regulator	
Information need	Notification only if a notifiable personal data breach occurs
Channel (primary / alternative)	Official channel
Owner	DPO with Principal approval
Timing	Within 72 hours of awareness
Media and public	
Information need	Holding statement
Channel (primary / alternative)	Approved statement
Owner	Comms Lead with Principal approval
Timing	On enquiry

> **Subject: Main site closed today - 11:00 examination postponed**
>
> Dear learner,
>
> Flooding and a power failure closed the Northbridge main site at 08:00 today, Monday. Please do not travel to the main site.
>
> Today's 11:00 examination will not take place, and you will not be penalised for this change. By 18:00 today we will send you the new date, time and venue, together with the arrangements for learners using their own devices.
>
> If you need a centre workstation, extra time or other support, please reply to this message or email learnersupport@northbridge.example by 14:00 today so that we can arrange your place.
>
> Fee payments made through our payment provider remain valid. We will send an online form for admissions and registration requests by 10:00.
>
> We will update you by email and text message each day until normal services return. We are sorry for the disruption and thank you for your patience while we restore our services.
>
> Learner Support Team, Northbridge Skills Institute (fictional)

4.6 Incident response trigger and controlled return

No cyberattack has been confirmed. The IR Lead is activated, while recovery continues under the IR Lead's direction for the affected components, if any of the following occurs: an unexpected administrator or backup-console logon, a backup integrity failure with no benign explanation, data changes or missing transactions that the checkpoint times do not explain, malware alerts, or a ransom demand (A12). The DPO then assesses whether a personal data breach is likely to put individuals' rights at risk. If it is, the regulator must be notified within 72 hours of awareness, and high-risk breaches must be communicated to the people affected (A1). [4, Sec. 40].

The return to the primary site follows NIST's reconstitution approach of validating data and functions before the plan is deactivated. Table 17 defines the controlled transfer of new records and final sign-off. [3, Ch. 4].

Table 17. Return to the primary site and final sign-off

Step	Criterion and evidence	Owner
1 Facilities clearance	Structural, electrical and flood-damage clearance certified in writing	Facilities Manager
2 Primary infrastructure ready	Power, network and servers tested; restored images verified	ICT Manager
3 Transaction freeze	Announced cut-off time; alternate systems set to read-only	BC/DR Coordinator
4 Transfer of new records	Final verified backup at the alternate site; restore to primary; record counts and checksums match	ICT Manager
5 Business validation	Each owner signs an acceptance checklist; temporary records fully reconciled	Function owners
6 Final sign-off and deactivation	Principal signs after steps 1-5; users notified; alternate copy kept read-only for 7 days (P6); review within 10 working days	Principal

Task 5 Recovery schedule and exercise

5.1 Baseline recovery schedule (modelled)

The recovery schedule uses the warm-site durations and two available technicians. Each action begins only after its prerequisites have finished, the assigned technician has no overlapping work, and no activity is split. The earliest technical restoration of LMS01 is 11:45. The controlling sequence is the recovery network (30 minutes), AUTH01 (30), security and backup checks (30), DB01 (90) and LMS01 (45), totalling 225 minutes. All five technical functions finish by 12:15. [1]

Table 18. Baseline recovery schedule (modelled, not a live test)

Field	Information
Recovery network	
Technician	T1
Start	08:00
Finish	08:30
Minutes	30
Prerequisite	None
AUTH01 restoration	
Technician	T1
Start	08:30
Finish	09:00
Minutes	30
Prerequisite	Recovery network
Security and backup checks	
Technician	T1

Field	Information
Start	09:00
Finish	09:30
Minutes	30
Prerequisite	AUTH01 restoration
DB01 restoration	
Technician	T1
Start	09:30
Finish	11:00
Minutes	90
Prerequisite	Security and backup checks
FS01 restoration	
Technician	T2
Start	09:30
Finish	10:30
Minutes	60
Prerequisite	AUTH01 restoration and security and backup checks
LMS01 restoration	
Technician	T1
Start	11:00
Finish	11:45
Minutes	45
Prerequisite	DB01 restoration
FIN01 restoration	
Technician	T2
Start	11:00
Finish	11:30
Minutes	30
Prerequisite	DB01 restoration
HR01 restoration	
Technician	T2
Start	11:30
Finish	12:15
Minutes	45
Prerequisite	DB01 restoration
REG01 restoration	
Technician	T1
Start	11:45
Finish	12:15
Minutes	30
Prerequisite	DB01 restoration

The 12:15 completion time is also a lower bound for this dependency structure: DB01 cannot finish before 11:00 because the shared prerequisite chain requires 180 minutes, and the four dependent application restores require 150 technician-minutes. Even with both technicians continuously occupied, those applications need another 75 minutes. The schedule attains this lower bound, as shown in Table 18 and Appendix C. Technician 2 is available for non-recovery preparation before FS01 can start at 09:30.

5.2 Comparison with objectives

Table 19 compares the baseline with the targets. It assumes 30 minutes for business acceptance (A8). All five functions meet their RTO and MTD. F1 has the smallest RTO margin, at 45 minutes. F1 to F4 meet their RPOs, but with no margin. F5 does not, because its 9-hour checkpoint exceeds its 4-hour RPO.

Table 19. Modelled baseline results against objectives (not a live test)

Field	Information
F1	
Technical finish	11:45
Elapsed from 08:00	3 h 45 min
Declared (+30 min, A8)	12:15 (4 h 15 min)
RTO / MTD	5 h / 24 h
Checkpoint age / RPO	1 h / 1 h
Result	Meets RTO, MTD and RPO; misses the 11:00 exam
F2	
Technical finish	11:30
Elapsed from 08:00	3 h 30 min
Declared (+30 min, A8)	12:00 (4 h)
RTO / MTD	5 h / 8 h
Checkpoint age / RPO	1 h / 1 h
Result	Meets all
F3	
Technical finish	12:15
Elapsed from 08:00	4 h 15 min
Declared (+30 min, A8)	12:45 (4 h 45 min)
RTO / MTD	6 h / 10 h
Checkpoint age / RPO	1 h / 1 h
Result	Meets all
F4	
Technical finish	12:15
Elapsed from 08:00	4 h 15 min
Declared (+30 min, A8)	12:45 (4 h 45 min)
RTO / MTD	12 h / 24 h
Checkpoint age / RPO	1 h / 4 h
Result	Meets all
F5	
Technical finish	10:30
Elapsed from 08:00	2 h 30 min
Declared (+30 min, A8)	11:00 (3 h)

Field	Information
RTO / MTD	24 h / 48 h
Checkpoint age / RPO	9 h / 4 h
Result	Meets RTO and MTD; misses RPO by 5 h

Several deadlines need a continuity workaround even when restoration succeeds. The 11:00 exam is 45 minutes earlier than the earliest LMS01 restoration. The seating shortfall remains whatever the restoration time. Confirmations under F2 before 12:00 rely on the provider portal. Online applications under F3 must be keyed in before the 18:00 close. Data from the 07:00 to 08:00 window, AUTH01 changes since 00:00 and FS01 changes since 23:00 on Day -1 must all be reconciled manually.

It is important to interpret the schedule narrowly. The two-technician model measures restoration activity and includes the local validation bundled in each duration; it does not establish that the building, supplier links or end-user devices will be available. A separate 30-minute owner acceptance allowance is applied consistently to each function, and those acceptances can proceed in parallel without occupying either technician. The baseline checkpoint assumption also matters: the 11:45 assessment finish is attainable only if the selected backup chain passes integrity and access checks. The failed-incremental inject is therefore a different, more severe case and must not be averaged into the baseline figures.

5.3 Tabletop exercise

The 45-minute tabletop exercise is documented as an individual role-based simulation. The first nine minutes establish the baseline and objectives. Injects are presented at minutes 10, 20, 30 and 40, with an initial decision recorded for each, followed by a closing review at minutes 43 to 45. The stress scenario is evaluated separately from the modelled baseline schedule. [1] [2, slide 11]; [3, Sec. 3.5].

Table 20. Exercise log (stress scenario, simulated roles)

Field	Information
10	
Inject	Closure extends from 72 hours to five days
Decision	Extend the alternate facility and recovery-site booking to 7 days; move the return date
Revised action	Facilities Manager confirms the extension; staff rota for 5 days; daily learner updates; F5 manual service extended past the 48-hour commitment
Owner	Facilities Manager
Rationale	Recovered services must be sustained, not only restored
Effect on service or recovery	No change to restoration times; continuity costs rise beyond the 24-hour estimate window
20	
Inject	Only 15 working workstations
Decision	Three sittings with three paper variants
Revised action	Day 1 09:00 and 14:00 sittings with 15 learners each; Day 2 09:00 sitting with 10 learners; adjustment learners in sitting 1; own-device learners unchanged
Owner	Head of Assessment
Rationale	Integrity is kept through variants; accessibility seats come first
Effect on service or recovery	The exam finishes one day later; staff lose spare workstations
30	
Inject	DB01 incremental fails its integrity check; unexpected backup administrator logon
Decision	Pause the DB01 restore; activate IR; preserve logs; disable the shared identity
Revised action	IR reviews the logon. If cleared, restore to the last verified incremental before the failure (A9). If not, restore from the Day -3 disconnected full (A10). The DPO assesses breach status
Owner	IR Lead (decision recorded by Principal)
Rationale	Restoring from a possibly tampered chain could spread compromise
Effect on service or recovery	RPO for F1-F4 breached by up to 8 h if only B01 (23:00) is usable, or by 62 h with the Day -3 fallback (Appendix C); RTOs slip by the length of the investigation; F2 and F3 continue manually
40	
Inject	Payment provider reports payments missing from restored finance records
Decision	Treat the provider export as the authoritative record of payments
Revised action	Match each missing payment by reference and time; post approved adjustments; learner confirmations stand; payments not explained by the checkpoint gap go to IR
Owner	Bursar
Rationale	Payments in the lost window are expected; unexplained gaps may indicate tampering
Effect on service or recovery	FIN01 accepted only after reconciliation; affected learners notified

The exercise showed that the plan's weakest points are its dependencies, not its timings. Injects 3 and 4 showed that one doubtful backup chain or administrator identity can turn a facilities outage into a possible security incident, and that the payment provider's records are the only independent check on finance data. Inject 2 showed that seating, rather than technology, limits the examination. The corrective actions in Table 21 target these findings, and each has evidence that can be checked.

Table 21. Corrective actions (proposals P10; target dates relative to fictional plan approval date P, A17)

Field	Information
CA1	
Action	Separate backup administration identities with MFA; remove the shared identity
Priority	High
Owner	ICT Manager
Target date	P + 30 days
Measurable completion evidence	Access review shows zero shared identities and 100% MFA enrolment
CA2	
Action	Add an immutable or offline third copy with a 35-day lock
Priority	High
Owner	ICT Manager
Target date	P + 60 days
Measurable completion evidence	Successful restore from the immutable copy; lock settings recorded
CA3	
Action	Move DB01 to 30-minute incrementals with automatic integrity checks and alerts
Priority	High
Owner	ICT Manager
Target date	P + 45 days
Measurable completion evidence	30 consecutive days of job reports with at least 99% verified
CA4	
Action	Obtain and test the offsite provider's access-independence controls
Priority	High
Owner	ICT Manager with Bursar
Target date	P + 60 days
Measurable completion evidence	Signed attestation and a witnessed test record
CA5	
Action	Contract surge capacity of at least 40 workstations, with a second facility option
Priority	Medium
Owner	Facilities Manager
Target date	P + 90 days
Measurable completion evidence	Signed addendum; walkthrough confirms 40 working workstations
CA6	
Action	Approve the exam contingency procedure with banked variants and a 09:00 go/no-go rule
Priority	High
Owner	Head of Assessment
Target date	P + 30 days
Measurable completion evidence	Approved procedure; two banked variants for every exam this term

Field	Information
CA7	
Action	Write a payment reconciliation procedure with daily provider exports
Priority	Medium
Owner	Bursar
Target date	P + 45 days
Measurable completion evidence	Test reconciliation completed with zero unexplained items
CA8	
Action	Cross-train a third recovery person
Priority	Medium
Owner	ICT Manager
Target date	P + 120 days
Measurable completion evidence	Supervised DB01 restore completed within the drill time

Task 6 Governance and performance measures

6.1 Ownership, testing and maintenance

Following the standard governance model, the BC/DR Coordinator owns and maintains the plan, and the Principal approves it on behalf of the Executive Steering Committee each year and after any major change. Function owners maintain their own procedures, and the ICT Manager maintains the runbook. Training is given at least annually, as NIST advises, and within 30 days for anyone newly appointed to a role (P7). Tables 22 and 23 set out the controls and the testing calendar. [2, slide 12]; [3, Sec. 3.6].

Table 22. Governance controls (proposals, P7)

Control	Proposal
Version control	Version number, change log and approval record; superseded copies withdrawn
Distribution	Controlled list of role holders; Microsoft 365 copy plus an encrypted offline copy; printed quick-reference cards at the alternate facility
Review triggers	Any activation or test; system, supplier or staff-role change; flood warnings; new legal requirements; annual review at a minimum
Training	Annual training for all role holders; induction within 30 days; records held by the HR Manager

Table 23. Risk-based testing calendar (proposals, P7; frequencies follow standard practice where stated)

Field	Information
DB01 restore test	
Timing	Monthly
Focus	Highest-impact shared data
Owner	ICT Manager
AUTH01 and FS01 restore tests	
Timing	Quarterly
Focus	RPO gaps closed
Owner	ICT Manager
Tabletop exercise	
Timing	Quarterly; one before each exam period
Focus	Exam and deadline decisions
Owner	BC/DR Coordinator
Walkthrough test	
Timing	Semi-annually
Focus	Runbook and temporary procedures
Owner	BC/DR Coordinator
Simulation with warm-site parallel test	
Timing	Annually, before the rainy season
Focus	Supplier timings and RTO evidence
Owner	ICT Manager
Call-tree test	
Timing	Quarterly

Field	Information
Focus	Alternative communication
Owner	Comms Lead

6.2 Framework mapping and climate consideration

Table 24 maps the continuity arrangements to the NIST seven-step method and to the ISO 22301 clause themes. The framework mapping supports internal alignment and does not imply certification. [2, slides 4 and 9]; [3]; [5].

Table 24. Framework mapping (educational alignment only)

Framework	Step or clause (standard wording)	Report output
NIST SP 800-34	01 Develop Contingency Planning Policy Statement	Ownership and approval (Section 6.1)
NIST SP 800-34	02 Conduct Business Impact Analysis (BIA)	Tables 1-3
NIST SP 800-34	03 Identify Preventive Controls	Backup design and access controls (Tables 6-7)
NIST SP 800-34	04 Create Contingency Strategies	Option matrix and budget (Tables 8-10)
NIST SP 800-34	05 Develop Information System Contingency Plan (ISCP)	DR runbook (Table 15)
NIST SP 800-34	06 Ensure Plan Testing, Training & Exercises (TT&E)	Schedule and tabletop (Tables 18-21)
NIST SP 800-34	07 Plan Maintenance	Governance and testing calendar (Tables 22-23)
ISO 22301	Clause 4 Context	BIA and climate consideration
ISO 22301	Clause 5 Leadership	Decision maker, deputy and risk acceptance (Tables 10-11)
ISO 22301	Clause 6 Planning	Recovery objectives (Table 4)
ISO 22301	Clause 7 Support	Training and communication (Tables 16, 22)
ISO 22301	Clause 8 Operation	Strategy, plan and temporary procedures (Tasks 3-4)
ISO 22301	Clause 9 Performance	KPI dashboard (Table 25)
ISO 22301	Clause 10 Improvement	Corrective actions (Table 21)

This disruption was caused by flooding. ISO's 2024 climate action amendment to ISO 22301 adds a requirement to clause 4.1 that the organisation determine whether climate change is a relevant issue, and a note to clause 4.2 that interested parties can have climate-related requirements. Northbridge should record climate change as relevant. In practice, the alternate facility, the warm site and the backup locations should sit outside the main site's flood area and power supply (A6), local servers should be moved above flood level, rainfall warnings should act as a pre-activation alert, and the annual simulation should run before the rainy season. NIST likewise lists the risk of the offsite storage site being hit by the same disaster as a selection criterion. [6].

6.3 KPIs and maturity

Table 25 defines the performance dashboard. Its targets are proposals (P8). The recovery, testing and compliance categories guide the choice of measures. K1 and K2 compare actual recovery with the targets in Table 4, K3 shows whether restore tests cover every critical system, K4 tracks the closure of exercise findings and K5 measures training. K6 adds the daily backup evidence on which every restore depends. [2, slide 13].

Table 25. Performance dashboard (targets are proposals, P8)

Field	Information
K1 Recovery time performance	
Formula or measurement method	Functions declared within RTO / functions tested x 100
Data source	Exercise and test logs
Target	100%
Frequency	Quarterly
Accountable owner	BC/DR Coordinator
Current baseline	5/5 modelled (not live)
K2 Recovery point performance	
Formula or measurement method	Systems whose restored checkpoint age is within RPO / systems tested x 100
Data source	Restore test records
Target	100%
Frequency	Monthly
Accountable owner	ICT Manager
Current baseline	1/3 (DB01 only; AUTH01, FS01 miss)
K3 Restoration-test coverage	
Formula or measurement method	Critical systems successfully restored in the period / 7 critical systems x 100
Data source	Restore test register
Target	100% each quarter
Frequency	Quarterly
Accountable owner	ICT Manager
Current baseline	Not yet measured
K4 Corrective-action closure	
Formula or measurement method	Actions closed by target date / actions due x 100
Data source	Corrective-action register
Target	At least 90%
Frequency	Monthly
Accountable owner	BC/DR Coordinator
Current baseline	0 of 8 due
K5 Training completion	
Formula or measurement method	Role holders trained in the last 12 months / all role holders x 100
Data source	HR training records
Target	100%
Frequency	Quarterly
Accountable owner	HR Manager
Current baseline	Not yet measured

Field	Information
K6 Backup integrity	
Formula or measurement method	Jobs passing integrity verification / all jobs x 100
Data source	Backup reports
Target	At least 99%
Frequency	Weekly
Accountable owner	ICT Manager
Current baseline	Not yet measured

On the maturity model used here, current maturity is Level 2, Developing (A15). Basic plans exist: a warm-site drill with timings and offsite backups. However, there is no evidence of regular testing, an identity is shared between production and backup, provider controls are undocumented and two RPO gaps remain, so Level 3 is not yet reached. The realistic 12-month target is Level 3, Defined: an approved plan, quarterly testing, KPIs reported to the Steering Committee and at least 90% of corrective actions closed. Level 4 is not realistic within a year, because it needs a measured performance history and a third technician. [2, slide 13].

Table 26. Maturity assessment and target

	Current (Level 2 Developing)	**12-month target (Level 3 Defined)**
	Warm-site drill and backups exist	Approved plan covering F1-F5 and controlled distributio
	No evidence of regular testing	Calendar in Table 23 delivered; K3 at 100%
	Shared identity; undocumented provider controls	CA1-CA4 closed; K4 at least 90%
	Not reported	Table 25 reported quarterly

References

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[5] International Organization for Standardization, "ISO 22301:2019 Security and resilience - Business continuity management systems - Requirements," standard overview page. [Online]. Available: https://www.iso.org/standard/75106.html

[6] International Organization for Standardization, "ISO 22301:2019/Amd 1:2024 Security and resilience - Business continuity management systems - Requirements - Amendment 1: Climate action changes," Feb. 2024, standard page. [Online]. Available: https://www.iso.org/standard/88412.html

Appendix A Assumptions register

Table A1. Assumptions register

ID	Assumption	Where used
A1	Northbridge operates in Nigeria (amounts in NGN, times in WAT), so the Nigeria Data Protection Act 2023 applies under s. 2(2)(a). The institute's location is unspecified; local legal applicability requires verification.	4.5, 4.6
A2	All owner roles, names and contact addresses are fictional; .example addresses are used.	Tables 1, 11-17
A3	The exam length is not stated; it is assumed to be no more than 3 hours, so two sittings (09:00 and 14:00) fit within Day 1.	4.3, Tables 14, 20
A4	A same-day delayed start is not treated as an acceptable assessment service by 11:00; the NGN 120,000 trigger is incurred.	4.3
A5	Split sittings with approved paper variants are acceptable under Northbridge's assessment rules (subject to Northbridge's assessment approval).	4.3, Tables 14, 20
A6	The locations of the alternate facility, warm site and offsite provider relative to the flood area and power supply are not stated. They are assumed to be separate, and Facilities must confirm this before activation. The provider's access-independence remains unverified.	2.3, Table 8, 6.2
A7	Own-device learners can reach the restored LMS01 remotely through the recovery network.	Table 14
A8	Business acceptance takes 30 minutes per function after technical completion and is done by function owners in parallel, not by technicians.	Table 19, Appendix C
A9	B02 incrementals run hourly from 00:00 to 07:00.	Table 20, Appendix C
A10	The 17:00 Day -3 disconnected full backup contains DB01; whether it also holds AUTH01 and FS01 is unconfirmed.	Tables 5, 15, 20
A11	The staff listed in Table 13 are available during the outage, on site or remotely.	Table 13
A12	An incident response capability exists: an internal IR Lead plus an external retainer.	4.6, Table 20
A13	An SMS and voice call tree from staff mobile phones is available as the alternative channel.	Tables 11, 16
A14	Business staff can work on laptops through Microsoft 365 and remote access, leaving workstations for learners.	4.3

ID	Assumption	Where used
A15	Documented test history is unavailable; maturity is assessed from the available operational evidence.	6.3
A16	Illustrative only: hourly costs stop when the minimum service is declared.	3.3, Appendix C
A17	Calendar dates are fictional; target dates are counted from a plan approval date, P.	Table 21

Appendix B Proposals and scenario data register

Table B1. My own proposals (targets, schedules, retention, thresholds)

ID	Proposal	Where used
P1	Recovery is declared at the Table 1 minimum service after owner acceptance (the usual definition measures RTO to full restoration).	2.1
P2	All RTO, RPO and MTD targets and buffers.	Table 4
P3	Backup frequencies, retention periods, placement and restore-test frequencies.	Tables 6-7
P4	Activation thresholds, 15-minute escalation and the NGN 250,000 escalation limit.	Table 11
P5	Minimum staffing numbers and the 4-working-hour acknowledgement for F5.	Table 13
P6	7-day read-only retention of the alternate copy; review within 10 working days.	Table 17
P7	Governance controls, 30-day induction training and testing items not specified in the standard model (monthly and quarterly restores, call-tree tests).	Tables 22-23
P8	All KPI targets.	Table 25
P9	The 09:00 go/no-go decision, the Day 1 sitting times and the paper variants.	4.3, Table 14
P10	Corrective actions, priorities and target dates.	Table 21

Table B2. Northbridge case data and guidance sources

Source	Inputs taken from it
Northbridge case dataset	Scenario, resources (25 workstations from 10:00, two technicians, 40 + 80 learners), hourly rates, NGN 120,000 trigger, deadlines, dependencies, backup checkpoints and weaknesses, package costs, estimates and additions, drill timings, injects
Course materials	BIA phases, RTO/RPO/MTD definitions and tiering, site types, backup types and 3-2-1, ISO clause themes, plan structure, test types and frequencies, governance and RACI, KPIs and maturity model, IR integration
Proposals	P1-P10 in Table B1 and the recommendation in Section 3.3

Appendix C Supporting calculations and schedule checks

Table C1. Disruption cost working (trigger met in both windows: 11:00 is 3 h after 08:00)

Function	8 hours (NGN)	24 hours (NGN)
F1	60,000 x 8 = 480,000 + 120,000 = 600,000	60,000 x 24 = 1,440,000 + 120,000 = 1,560,000
F2	40,000 x 8 = 320,000	40,000 x 24 = 960,000
F3	20,000 x 8 = 160,000	20,000 x 24 = 480,000
F4	10,000 x 8 = 80,000	10,000 x 24 = 240,000
F5	5,000 x 8 = 40,000	5,000 x 24 = 120,000
Total	1,200,000	3,360,000

Table C2. Checkpoint ages and inject 3 fallback (reference point 08:00 Day 0)

Field	Information
DB01 (B01 + B02 to 07:00)	
Working	08:00 - 07:00
Age (h)	1
Gap against RPO	1 - 1 = 0 (F1-F3)
AUTH01 (B03)	
Working	08:00 - 00:00
Age (h)	8
Gap against RPO	8 - 4 = 4 over
FS01 (B04)	
Working	1 h to midnight + 8 h
Age (h)	9
Gap against RPO	9 - 4 = 5 over
DB01 B01 only (inject 3, first incremental fails, A9)	
Working	1 h to midnight + 8 h
Age (h)	9
Gap against RPO	9 - 1 = 8 over
Disconnected full, 17:00 Day -3	
Working	7 h + 48 h + 8 h
Age (h)	63
Gap against RPO	63 - 1 = 62 over
Proposed DB01 chain (Table 6)	
Working	Sunday 23:00 full + 17 incrementals 23:30-07:30; 08:00 - 07:30
Age (h)	0.5
Gap against RPO	1 - 0.5 = 0.5 margin

Table C3. Budget working (NGN)

Field	Information
Cold	
Working	600,000 + 400,000 + 200,000

Field	Information
Total	1,200,000
3,000,000 - total	1,800,000
Warm	
Working	1,500,000 + 400,000 + 200,000
Total	2,100,000
3,000,000 - total	900,000
Hot	
Working	2,400,000 + 400,000 + 200,000
Total	3,000,000
3,000,000 - total	0

Table C4. Schedule verification

Check	Result
Earliest DB01 finish	30 + 30 + 30 + 90 = 180 min, so 11:00
Lower bound for LMS01 finish	180 + 45 = 225 min, so 11:45
Lower bound for all functions	180 + (45 + 30 + 30 + 45) / 2 = 255 min, so 12:15
Schedule feasibility	The proposed parallel allocation reaches the 255-minute lower bound without technician overlap, pre-emptions or unmet prerequisites.
Table 18 schedule	All prerequisites finish before each start; no technician overlap; no split actions; reaches both lower bounds
Exam gap	LMS01 finishes 45 min after the 11:00 exam start

Table C5. Elapsed and declared times (declared = technical finish + 30 min, A8)

Field	Information
F1	
Technical finish	11:45
Elapsed (h)	3.75
Declared (h)	4.25
RTO margin (h)	0.75
F2	
Technical finish	11:30
Elapsed (h)	3.50
Declared (h)	4.00
RTO margin (h)	1.00
F3	
Technical finish	12:15
Elapsed (h)	4.25
Declared (h)	4.75
RTO margin (h)	1.25

Field	Information
F4	
Technical finish	12:15
Elapsed (h)	4.25
Declared (h)	4.75
RTO margin (h)	7.25
F5	
Technical finish	10:30
Elapsed (h)	2.50
Declared (h)	3.00
RTO margin (h)	21.00

Table C6. Illustrative modelled cost with the warm baseline (A16)

Function	Working	Cost NGN
F1	60,000 x 4.25 + 120,000	375,000
F2	40,000 x 4.00	160,000
F3	20,000 x 4.75	95,000
F4	10,000 x 4.75	47,500
F5	5,000 x 3.00	15,000
Total	Sum	692,500
Avoided against 24-hour estimate	3,360,000 - 692,500	2,667,500
Warm annual cost as share of 24-hour estimate	2,100,000 / 3,360,000	62.5%